

GE23311/GE23411 - ENVIRONMENTAL SCIENCE AND SUSTAINABILITY

CLASS: II Year All Branches

UNIT II – ENVIRONMENTAL POLLUTION

Syllabus: Causes, effects and preventive measures of water, soil, air and noise pollution. Types of wastes and E-waste management. Case studies on Occupational Health and Safety Management system (OHSAS). Environmental protection, Environmental protection acts.

Pollution:

The environmental pollution may be defined as the unfavourable alteration of our surrounding. It changes the quality of air, water and land.

Pollutants:

The substances which adversely alter the quality of the natural environment are called pollutants.

- **Biodegradable pollutants** – Can be broken down naturally.
Examples: sewage, paper, food waste.
- **Non-biodegradable pollutants** – Persist in the environment for a long time.
Examples: plastics, pesticides (DDT), heavy metals (mercury, lead).

Types of pollution:

1. Water pollution
2. Soil pollution
3. Air pollution
4. Noise pollution

Causes for pollution:

- Population explosion
- Unplanned urbanization
- Industrialization
- Deforestation
- Excessive heat, noise, light or radiation from industrial sectors.

AIR POLLUTION

Definition:

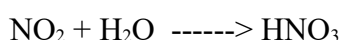
The presence of one or more contaminants like dust, smoke, mist & odour in the atmosphere, which are harmful to living beings, plants & animals.

Types of Air pollution

Primary Air Pollutant: Emitted directly in the atmosphere in the harmful form.

Ex. CO₂, NO, CO

Secondary Air Pollutant: Primary air pollutant may react with one another or with basic component of air.



Indoor air pollutants are primary air pollutants. Ex. Radon gas emitted from building materials, burning of fuels in Kitchen.

Cause of Air pollution

- Urbanization
- Population
- Deforestation
- Industrialization
- Vehicle emission

Sources and Effects of Air Pollution:

S.No.	Pollutant	Description	Human Sources	Health Effects	Environmental Effects
1	CO	Colourless, Odourless and Poisonous $2\text{C} + \text{O}_2 \longrightarrow 2\text{CO}$	Cigar Smoking, Incomplete burning of fossil fuels	Reacts with blood cell and causes head ache and anaemia.	It increases the global temperature
2	NO ₂	Reddish Brown irritating gas $\text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_3$	Motor vehicle emission, Fossil fuel burning	Lung Irritation and damage.	Damages tree, Soils. It affects the aquatic life.
3	SO ₂	Colourless, Irritating gas	Coal burning, Power plants	Breathing problem for health people	Reduce visibility, It causes Acid rain.

4	O ₃	Highly reactive, irritating gas, unpleasant odour	Chemical reaction. With volatile org. compounds and NO.	Rapid Heartbeat, chest pain	Moderate the Climate
5.	Hydrocarbon	Due to decay of vegetable matter	Agriculture , decay of plants	Carcinogenic	Until it reacts with other compounds to form sec. pollutant, there is no problem.
6	Suspended Particulate Matter	Particles and droplets suspended in atmosphere.	40% - Industries, 17% - Automobiles	asthma, cancer	Reduce visibility , damage aquatic life
7	Photochemical Smog	Unburnt Hydrocarbon + NO _x -----> Smog	Burning of coal, power plant.	Throat irritation, heart diseases.	Reduce visibility, damage plants and trees
8	Lead	Solid toxic metal.	Paint, petrol. leaded	affects brain, nervous system,	can harm wild life
9	Chromium	Solid toxic metal.	Paint, chromium plating.	lung cancer, nasal, skin ulcers,	toxic, carcinogenic, major environmental problem

Control Measures:

Source Control:

- Plant more trees.
- Use solar battery in automobiles.
- Use unleaded petrol.
- Industries should be situated outside.
- Houses, schools are not located on busy streets.

Industry level:

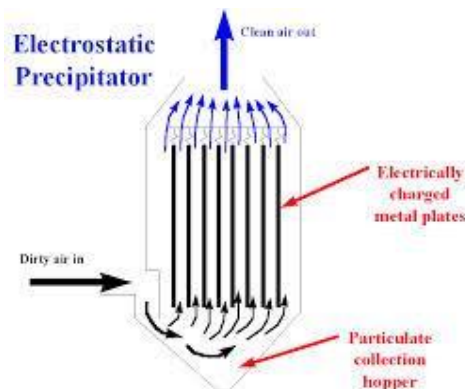
- Emission rate should be controlled. Use air pollution control equipment's. Continuous monitoring of pollutants

Equipment used to control air pollution:

- To ensure sufficient supply of oxygen to combustion chamber.
- Adequate temperature for combustion
- Mechanical device like scrubbers, cyclones, bag houses and electrostatic precipitators.

Techniques used for prevention of Air pollution

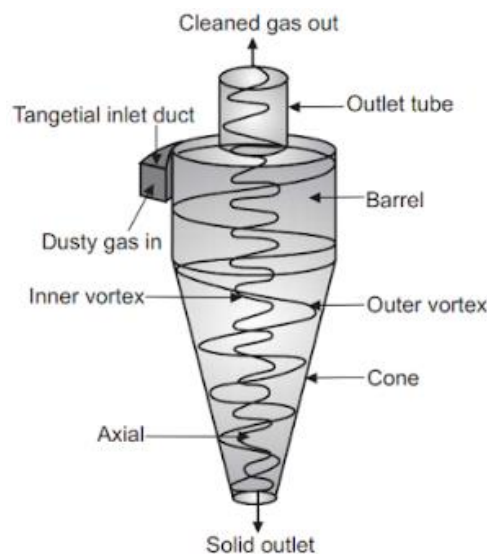
1. **Equipment used to control the air pollution:** The atmosphere has several built in self-cleaning process such as dispersion, gravitational settling, flocculation and absorption so on, to cleanse the atmosphere. Some of the equipment's are used to remove the air pollutant from the source.
- **Electrostatic Precipitator:** The impure air is passed to the chamber maintained at a high electrostatic potential of about 30000 v. Under the influence of higher potential the negatively charged particles of dust are attracted towards to the positively charged electrode. Then the hot dust free air is left out at the top of the chamber. The dust particles are collected in the hopper provided in the bottom.



- **Baghouse filter:** The impure air is filtered, when the air is passed through the baghouse filter. The filter consists of cellulose, nylon materials to absorb the dust particles. Then the bags are shaken out to get the dust particles in the bottom.



- **Filters:** filter remove particulate matter from the gas stream. Bag house filter system is the most common and it is made up of cotton fibres. When polluted gas passed through it, then polluted gas are deposited on cotton fibres.
- **Electrostatic scrubber:** the emitting dust is charged with ions and ionized particulate matter is collected on oppositely charged surface. The collected particles are removed by shaking the surface.
- **Scrubbers:** Scrubbers are wet collectors. They remove aerosol from a stream of gas either by collecting wet particle on a surface followed by their removal on the particles are wetted by scrubber liquid.
- **Cyclone separators** are devices that use centrifugal force to separate particles from a gas stream. They are commonly used in various industries to remove particulate matter from air and gas streams, improving air quality and preventing damage to downstream equipment.



Case Studies:

1. Air Pollution:

(i) The Taj Mahal's Marble Turning Yellow (Acid Rain)

Location: Agra, Uttar Pradesh

Over the past two decades, the iconic Taj Mahal has suffered discoloration. Its pristine white marble turning yellow due to air pollution. Emissions from nearby industries including rubber processing, chemical plants, and the Mathura oil refinery release sulphur dioxide (SO_2) and nitrogen dioxide (NO_2).

These gases combine with atmospheric water vapor to form sulphuric and nitric acids, leading to acid rain, which corrodes the marble by the phenomenon often, termed “Marble cancer.” Suspended particulate matter (e.g., soot) further adds to the yellowing effect.

Significance

- Highlights how air pollution doesn't just harm human health but also damages cultural heritage.
- Demonstrates the far-reaching effects of industrial emissions on iconic monuments.



Taj Mahal – Before and After Effects of Acid Rain

(ii) Delhi Air Pollution

Delhi consistently ranks among the world's most polluted cities, with $\text{PM}_{2.5}$ and NO_2 concentrations far exceeding health guidelines. In 2019, Delhi's annual average $\text{PM}_{2.5}$ concentration stood at $98.6 \mu\text{g}/\text{m}^3$, more than four times the World Health Organization's recommended limit. In earlier decades, total suspended particulate (TSP) levels were about five times the WHO's annual standard, exceeding unsafe thresholds on 97% of recorded days.

Primary Pollutants

- Vehicular emissions account for the largest share for about approximately 67% of air pollution, followed by coal-based power plants (~12%) and small-scale industries (~12%)PMC.
- Seasonal crop residue burning in neighboring states contributes up to 40% of Delhi's $\text{PM}_{2.5}$ load during post-monsoon and early winter.

Health Impacts & Human Cost

- Air pollution in Delhi is linked to 10,000–30,000 premature deaths annually due to cardiovascular and respiratory illnesses.

- The heavy burden of pollutants disproportionately affects vulnerable populations, leading to more life-years lost due to earlier onset of deaths compared to other regions.
- A landmark longitudinal study found that long-term exposure to PM_{2.5} increases the risk of type 2 diabetes, based on 12,000 individuals from Delhi and Chennai (2010–2017)



Delhi's Air Pollution

WATER POLLUTION

The alteration in physical, chemical and biological characteristics of water which cause harmful effects on humans and aquatic life.

Source of water pollution

Point and Non-Point sources of water pollution

- Point Sources-** These are discharged pollutants at specific locations through pipes, sewers in to bodies of surface water.

Example – factories, sewage treatment plants

- Non-point Sources-** They are usually large land areas that pollute water by run-off .location of which cannot be easily identified.

Example – Acid deposition, run-off of chemicals in to surface water from croplands, urban street etc.

Types, sources and effects of water pollution

S.NO	TYPE OF POLLUTANT	HUMAN SOURCES	EFFECTS
1	Infectious Agent	Human and animal	Variety of diseases
2	Dissolved Oxygen	Sewage, paper mills	Bacteria's are decomposing these wastes can degrade water quality by depleting do
3	Inorganic Chemicals (Water Soluble Acid, Pb, NaCl)	Surface runoff, industrial effluents	Unsuitable for drinking, damage nervous, liver and kidney
4	Organic Chemicals - Oils, Plastics And Pesticides	Surface runoff, industrial effluents	Harm fishes and wild life, damages the nervous system
5	Sediments (Soil)	Land erosion	Carry pesticides, bacteria can reduce
6	Radioactive Isotope (Iodine, U, Ce)	Nuclear power plants	Genetic mutation, birth defects
7	Excessive Heat	Industrial plants	Lower the DO, aquatics
8	Point Sources	Discharged pollutants at specific location through pipes, ditches etc.	
9	Non-Point Sources	Large land areas are polluted by water runoff. Location cannot be identified.	

Control measure for preventing water pollution

- Industrial effluent and domestic waste must be treated before disposal.
- Recycling of waste water through waste water treatment.
- Public awareness program.

Domestic Sources

- Proper sewage treatment before discharge.
- Use of eco-friendly detergents and household cleaners.

- Avoiding direct disposal of waste, plastics, and oils into water bodies.

Industrial Sources

- Installation of effluent treatment plants (ETPs).
- Adopting cleaner production techniques (reuse & recycling of water).
- Regular monitoring of wastewater quality before discharge.
- Shifting to less polluting raw materials and processes.

Agricultural Sources

- Controlled use of fertilizers and pesticides (Integrated Pest Management).
- Practicing organic farming methods.
- Constructing buffer zones and vegetative cover near water bodies to reduce runoff.
- Adopting drip irrigation to reduce fertilizer leaching.

Municipal and Urban Measures

- Proper solid waste management systems to avoid leachate pollution.
- Rainwater harvesting and groundwater recharge.
- Strict enforcement of laws (Water Act, 1974 in India).
- Public awareness campaigns for water conservation and cleanliness.

Ecosystem & Community Measures

- Afforestation along riverbanks to prevent erosion and siltation.
- Protection of wetlands which act as natural filters.
- Community participation in river and lake cleaning drives.
- Promoting the principle of Reduce, Reuse, Recycle.

Case Studies:

Water Pollution: River Ganga Severely Contaminated

Location: River Ganga (especially stretches like Varanasi, Haridwar)

The Ganga is facing escalating pollution from multiple sources:

Daily disposal of nearly 89 million litres of untreated sewage from towns along the river route.

- During high pilgrimage seasons such as the Char Dham Yatra, sewage inflow intensifies.
- Other contributors include partially burnt human remains and hazardous medical waste.
- Biological contamination is alarming: in Varanasi, coliform bacterial levels are up to 3,000 times higher than WHO safety standards.

Significance

- Illustrates how urbanization, religious practices, and infrastructural deficits pose grave health and environmental risks.

- Emphasizes the dire need for better sewage treatment, waste management, and public awareness.

The Ganga River is severely polluted due to untreated sewage, religious activities, and medical waste, especially in cities like Varanasi and Haridwar.

During peak pilgrimage seasons, pollution levels increase further.

High coliform bacteria levels show serious health and environmental risks, highlighting the need for better waste management.



The Ganga Pollution Problem

SOIL POLLUTION

- Soil Pollution can be defined as “The contamination of soil by human and natural activities which may cause harmful effects on living beings”

Types, effects and sources of soil pollution:

S.No.	Pollutant	Sources	Effects
1	Industrial wastes	Paper mills, Sugar mill, Oil refineries, Fertilizer industry etc.	Chemical and biological properties of the soil are affected.
2	Urban wastes	Plastics, Glasses, metal cans, rubber containers etc.	They cannot be degraded easily. It affects the soil fertility.
3	Agricultural practices	Fertilizer, Pesticide, Herbicide and weedicides etc.	It also affects the soil fertility.
4	Radioactive Pollutant	Nuclear testing industries U, Th, Ra etc.	It penetrates the soil and creating land pollution.
5	Biological wastes	Human, animals, Birds excretion.	Digested Sludges causes serious Damage to plants within a few years.

Control measures of soil pollution:

Control of soil erosion:

- Trees may planted on barren areas
- Reduce deforestation
- Maintain vital nutrient in soil.

Proper dumping of unwanted materials

- Open dumping is the common practice
- Recently tipping is followed for solid waste disposal

Production of natural fertilizer

- Use of excessive addition of pesticides to be avoided
- Used of bio pesticides to be encouraged.

Public Awareness

- Formal and informal talk should be given by the government
- To educate the people about the hazard in the environment.

Recycling and reuse of water

- Wastes such as paper, plastics, metals and glass should be recycled and reused.

Ban Toxic chemicals

- Banned chemicals should not be used. DDT, BHC are fatal to animals and plants.
Radioactive wastes should be banned.

Case Studies:

Soil Pollution: Heavy Metal Contamination in Manali Industrial Area, Chennai

Location: Manali industrial zone, Chennai, Tamil Nadu

The industrial area like Manali contains petrochemical units, refineries, and fertilizer plants. The soil has been found to contain dangerously high levels of several heavy metals. Samples revealed concentrations of chromium (Cr) up to 418 mg/kg, copper (Cu) as high as 372 mg/kg, nickel (Ni) up to 78.8 mg/kg, zinc (Zn) up to 213.6 mg/kg, and molybdenum (Mo) between 2.3–15.3 mg/kg. These levels significantly exceed typical safe thresholds, indicating severe soil degradation

The Manali industrial zone in Chennai is facing severe soil pollution due to petrochemical units, refineries, and fertilizer plants. Soil samples show dangerously high levels of heavy metals like chromium, copper, nickel, zinc, and molybdenum, exceeding safe limits. This indicates serious soil degradation and poses risks to both human health and the environment.



Significance

- Agricultural & Health Risk: Contamination can enter the food chain via crops or dust, posing long-term risks.
- Industrial Impact: Shows how unchecked industrial waste leads to persistent environmental hazards.

NOISE POLLUTION

“Unwanted, Unpleasant or disagreeable sound that causes discomfort for all living beings”

Types of Noise Pollution

Industrial noise

- Noise pollution is caused by the machines.
- Noise from Mechanical saw and pneumatic drills (tool to break up rocks) is unbearable and nuisance to public.
- In the steel industry, the workers near the heavy industrial blowers are exposed to 112 dB for eight hours and suffer from occupation pollution.

Transport Noise

- Road traffic noise, rail traffic noise and air craft noise are example for transport noise.
- Transport noise is increasing more in the cities because of increasing crowd.
- According to the experts, the noise level in the most of the residential areas in metropolitan cities is already hovering (on the edge) on the border line because of vehicular noise pollution.
- A survey shows that metropolitan cities like Delhi, Mumbai, Kolkata and Chennai has a noise level of 90dB, which can make people deaf.

Neighbourhood noise:

- This type of noise includes disturbance from household gadgets. TV, VCR, load speakers, Radio, transistor etc.,
- Example for household gadgets which creates noise pollution.

Causes of Noise Pollution:

- a) **Urbanization and Industrialization:** Rapid urbanization and industrial activities contribute significantly to noise pollution due to construction, traffic, factories, and machinery.
- b) **Transportation:** Vehicles such as cars, trucks, buses, airplanes, and trains produce high levels of noise, especially in densely populated areas or near transportation hubs.
- c) **Construction Activities:** Construction sites generate loud noises from machinery, drilling, hammering, and other activities, particularly during daytime hours.
- d) **Social Events and Entertainment:** Loud music from concerts, clubs, parties, and other social events can contribute to noise pollution, especially if they occur late at night or in residential areas.
- e) **Technological Devices:** Electronic gadgets, household appliances, and industrial equipment can emit noise pollution during operation.
- f) **Natural Phenomena:** Natural events like thunderstorms, earthquakes, and volcanic eruptions can also produce loud sounds, although these are typically temporary and less frequent contributors to noise pollution.

Effects of noise pollution

- Human health gets affected. It causes high blood pressure, mental stress, heart attack, birth defects and abortion.
- It causes nervous breakdown and high tension.
- Noise pollution changes the hormone content in blood and dilation of pupil.
- It damages brain, kidney liver and may cause emotional disturbance.
- Noise pollution may affect the ear drum and damage the auditory system.
- It causes psychological and pathological disorders.
- Recently it is reported that blood is also thickened by excessive noise.

Effects of Noise Pollution Auditory effects

- **Auditory fatigue** -- Whistling & buzzing in ears(noise level -90dB)
- **Deafness** -- Permanent hearing loss (noise level-100dB)

Non auditory effects

- Interference with speech communication -50dB
- Annoyance, ill temper, bickering
- loss in working efficiency - tiredness, deterioration or complete loss of ability to work

Physiological disorders

- Neurosis, anxiety, hypertension, increased sweating, giddiness, nausea, fatigue, visual disturbance, reduces depth and quality of sleep , peptic ulcers, Increases cholesterol level resulting in constriction of blood vessel , Low weight children to mothers

Control Measures

- **Source Control:** This may include source **modification** such as **acoustic treatment** to **machine** surface, design changes.
- **Transmission path intervention:** This may include keeping source in the **sound insulating box**, construction of noise barrier or using sound absorbing substance along the path of the sound.
- **Receptor Control:** This includes protection of the receiver by altering the work schedule or using personal protection devices such ear plugs.
- **Oiling:** Proper oiling can reduce the noise from the machine.
- **Planting more trees** which can act as effective **noise barrier**.

Case Studies:

Chennai – Persistent Urban Noise-Tamil Nadu

- **Citizen Matters (2022):** Chennai ranks highest among Indian metros for noise pollution—driven by traffic and construction. Silent zones near sensitive locations like schools and hospitals also suffer from excessive noise levels.
- **Recent Monitoring (2025):**
 - T Nagar registers over 65 dB at night, exceeding TNPCB's limit of 55 dB.
 - Silent zones in Egmore and Anna Nagar routinely record 60–65 dB, breaching the 50 dB cap.

In Chennai, areas like T. Nagar, Egmore, and Anna Nagar record night-time noise levels above permitted limits due to traffic and construction, even in silent zones near schools and hospitals. Similarly, in Mumbai, studies show commercial and residential areas consistently exceed CPCB noise standards, with serious health impacts such as hearing loss reported among firemen at KEM Hospital. These cases highlight the urgent need for stronger enforcement and noise-control measures in Indian cities.

- Neighbourhoods like Sow carpet, Velachery, and Triplicane show widespread violations.
- Main sources: vehicular noise and unregulated construction.
- Mitigation suggestions include noise-reducing infrastructure—double-paned windows, sound barriers, noise-absorbing materials.
- A noise-mapping initiative by TNPCB in partnership with IIT-Madras has been launched to tackle the problem, though enforcement remains limited.

1. **Mumbai-Ambient Urban Noise Across Zones- Maharashtra**

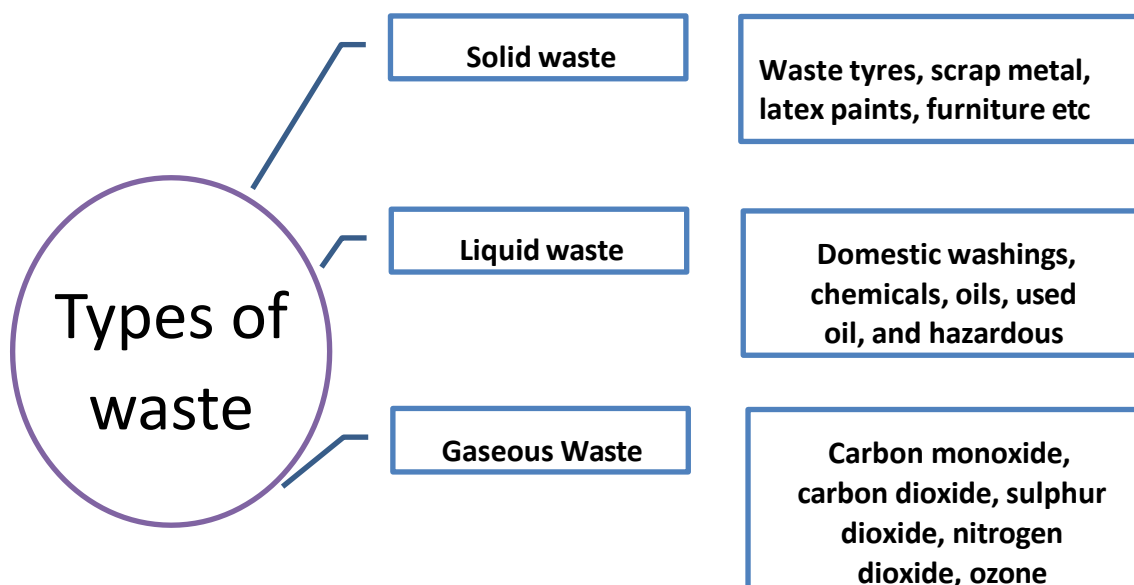
A more recent study (possibly by National Environmental Engineering Research Institute - NEERI) monitored 153 locations over 48 hours covering commercial, residential, industrial, and silence zones. Key findings:

- **Commercial areas:** Daytime levels between 75–90 dB; night time still at 70–80 dB far above the 65 dB CPCB limit.
- **Residential zones:** All readings exceeded the day limit of 55 dB and night limit of 45 dB.
 - 81% sites registered 75–85 dB during weekdays.
 - Weekend nights saw noise rise to 65–80 dB, often peaking at 95 dB.

Health impact: A KEM Hospital study found that 37.5% of Mumbai's firemen (ages 45–60) had noise-induced hearing loss.

Waste

- Wastes are unwanted or unusable materials. Waste is any substance which is discarded after primary use, or is worthless, defective and of no use.



SOLID WASTE MANAGEMENT OR SOIL WASTE MANAGEMENT

- Any material that is thrown away or discarded as unwanted is considered as solid waste. Management of the solid waste is very important in order to minimize the adverse effect of solid waste.

Objective of Solid Waste Management

- The primary objective of solid waste management is reducing and eliminating adverse impacts of waste materials on human health and environment.
- To control, collect, process, dispose of solid wastes in an economical way consistent with the public health protection.
- To support economic development and superior quality of life.
- Identification of Waste and its Minimization at the Source
- Collection, Segregation and Storage at the Site of Collection
- Safe transportation of Solid Waste
- Proper treatment of Solid Waste
- Energy recovery and effective disposal of solid waste.

Types and sources of solid waste:

There are 3 types of solid waste

1. Urban Waste
2. Industrial Waste
3. Hazardous Waste

1. Sources of Urban waste:

- ✓ Domestic waste – these are thrown out from the homes.
Example: Food waste, cloth, waste paper, glass bottles etc.
- ✓ Commercial waste – the waste coming out from shops, markets, hotels, offices etc.
Example: Waste paper, packing material, bottles etc.
- ✓ Construction waste - waste of construction materials
Example: Wood, Concrete, debris etc.
- ✓ Biomedical waste – waste organic materials
Example: Infectious wastes.

2. Sources of Industrial waste:

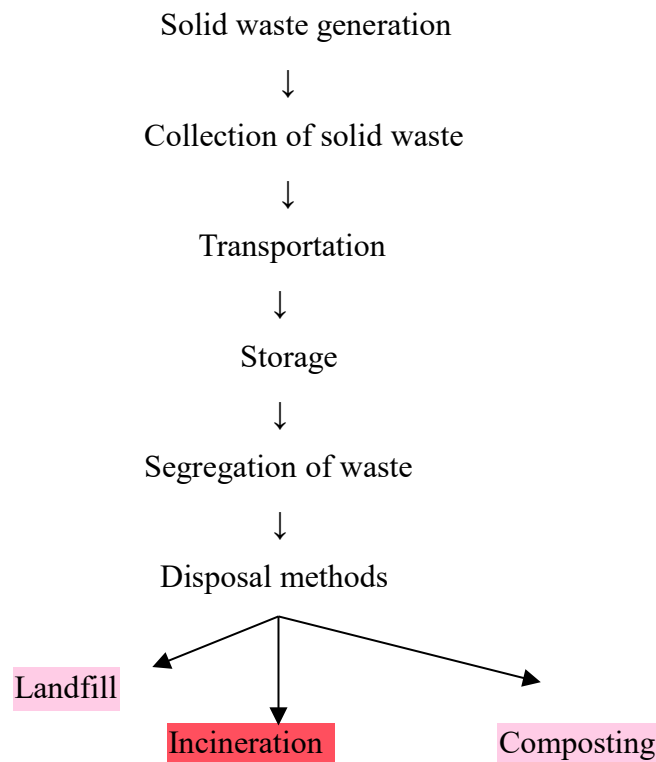
- ✓ Nuclear power plants – radioactive wastes
- ✓ Thermal power plants – produce fly ash

- ✓ Chemical Industries – hazardous and toxic materials
- ✓ Other industries – packing materials, organic waste, acid, alkali etc.

3. Sources of hazardous waste:

- ✓ Petroleum refineries, chemical manufacturing companies, paper mills, smelters, radioactive substances etc.

PROCESS OF SOLID WASTE MANAGEMENT



Steps involved in solid waste management

Reduce Reuse and Recycle

- **Reduce:** If the usage of raw materials reduced the generation of waste also gets reduced
- **Reuse:** The refillable containers, cycle tube (used for the production of rubber rings) can reuse.
- **Recycle:** discarded material converted in to new useful products and bottles.

Eg. Old aluminium bottles and glass bottles are melted and recast into new cans

Discarding waste

Methods of disposal of solid waste

1. Landfill
2. Incineration
3. Composting

1. Landfill

- In landfills solid wastes are filled alternatively – 80 cm solid waste covered with 20 cm earth fill
- After 2 or 3 years the solid waste volume shrinks by 25 to 30%.
- This land can be used for parks, roads and small buildings.

Advantages

- Simple and economical
- segregation not required
- converts low-lying areas in to useful lands

Disadvantages

- large area is required
- Land is away from City, transportation cost is high
- Bad odour- if land is not properly managed
- Landfill areas are the sources of mosquitoes and flies

2. Incineration

- It is a hygienic way of disposing solid waste
- In this method solid waste are burnt in a furnace called incinerator.
- The non-combustible materials can be left out for recycling and reuse
- The left out ashes is only about 10 to 20%
- The heat produced is used for generating electricity
- Incinerate about 100 to 150 tonnes per hour, Chamber temperature maintained between 700 to 1000°C

Advantages

- It requires very little space
- cost of transportation is not high because they are located within city limits.
- Safest from hygienic point of view
- 300 tonnes of waste used for the production of 3MW of power.

Disadvantages

- Capital and operating cost is high
- Needs skilled personnel
- Formation of dust, smoke causes air pollution.

3. Composting

- Compostable waste dumped in earthen trenches in layers of 1.5m and covered with earth of about 20 cm and left for decomposition
- For active decomposition microorganism (actinimycetes) introduced.
- Within 2 to 3 days biological action starts, Lot of heat is liberated(75°C)
- Finally the refuse converted in to powdery brown colour odourless mass known as humus used in agricultural field.

Advantages

- A number of industrial solid waste can be treated by this method
- The manure can be sold
- Recycling occurs

Disadvantages

- The non-consumables have to be disposed separately

Hazardous waste

It is the waste that has potential threats to public health (or) the environment.

Examples

- ✓ Cleaning solvents (acids and bases).
- ✓ Spent acids and bases.
- ✓ Metal finishing wastes.
- ✓ Painting wastes.
- ✓ Sludges from air and water pollution control units.
- ✓ Disinfectants and pesticides.

Types and characteristics of hazardous wastes

Toxic wastes

These are poisonous even in very small (or) trace amounts. They may have

(i) Acute effects

Causing death (or) violent illness

(ii) Chronic effects

Slowly causing irreparable harm.

Carcinogenic waste

It causes cancer after many years of exposure.

Mutagenic

- It causes major biological changes in the off-spring of exposed humans and wild life.

Reactive wastes

- These are chemically unusable and react violently with air (or) water.
- They cause explosions (or) form toxic vapours.

Ignitable wastes

- They burn at relatively low temperatures and cause an immediate fire hazard.

Corrosive wastes

- These include strong acidic (or) alkaline substances.
- They destroy solid material and living tissue upon contact.

Hazardous waste management

Definition

It is the collection, treatment and disposal of waste materials that can cause substantial harm to human health (or) to the environment.

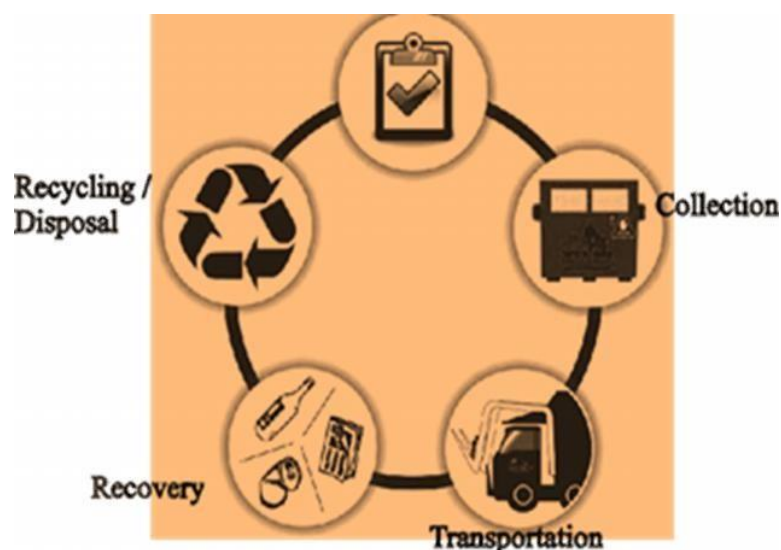
Improper hazardous-waste storage (or) disposal contaminates surface water and ground water supplies as harmful water pollution and land pollution.

People living in homes, built near waste disposal sites, may be in a vulnerable position.

The best remedy for this problem is to regulate the practice of hazardous - waste management.

Various steps of hazardous waste management

Hazardous waste management involves the following 4 steps



Step 1: Analysis:

- ✓ Physical and chemical properties of hazardous waste must be analysed before collection and recovery of useful components.
- ✓ It is essential because it can be used as a fertilizer, liming material (or) soil amendment.

Step 2: Collection and transport

- ✓ Hazardous waste, generated at a particular place, is generally collected and transported by truck over public highways.
- ✓ It can also be shipped in tank trucks, made of steel (or) aluminium alloy, with capacities upto about 34,000 litres.
- ✓ It can be containerized and shipped in 200 litre drums.

Step 3: Treatment (or) Recovery

Hazardous waste can be treated (or) recovered by

- ✓ Chemical method.
- ✓ Thermal method.
- ✓ Biological method.
- ✓ Physical method.

Bio-remediation

- ✓ Microbes can also be used for stabilizing hazardous wastes on previously contaminated sites.
- ✓ This process is called bio-remediation.

Step 4 Storage and disposal

- ✓ Hazardous wastes that are not destroyed by incineration (or) other chemical processes need to be disposed properly.
- ✓ This can be done by the following methods.

E-Waste

- Electronic waste, "e-waste" or "Waste Electrical and Electronic Equipment."
- Waste consisting of any Unwanted, obsolete or unusable electrical or electronic appliance.
- It is a point of concern considering that many components of such equipment are considered toxic and are not biodegradable.

Example: computers, mobile phones, televisions, home appliances etc..

SOURCES AND CAUSES OF E-WASTE:

- IT & Telecom Equipment's
 - Large Household Appliances
 - Small Household Appliances
 - Consumer & Lighting Equipment's
 - Electrical & Electronic Tools
 - Toys, Leisure & Sports
 - Equipment
 - Medical Devices
 - Monitoring & Control Instruments
- i) **Rapid Technological Advancement:** The ever-accelerating pace of technological innovation leads to frequent upgrades and replacements of electronic devices, contributing to a high turnover rate of electronic products.
 - ii) **Short Product Lifespan:** Planned obsolescence and a focus on producing cheaper, disposable electronics rather than durable goods shorten the lifespan of electronic devices, leading to more frequent disposal.
 - iii) **Consumerism and Disposal Culture:** Consumer behavior, driven by trends in fashion and technology, encourages the constant upgrading and discarding of electronics, even when they are still functional.
 - iv) **Globalization of Production:** The global supply chain for electronics often involves manufacturing in countries with less stringent environmental regulations, leading to increased environmental impact during production and disposal phases.

Effects of E-Waste:

Environmental Pollution: Improper disposal of e-waste leads to the release of hazardous substances such as lead, mercury, cadmium, and brominated flame retardants into the environment, contaminating soil, water, and air.

Health Risks: Exposure to toxic chemicals from e-waste, whether through direct contact, inhalation of fumes from burning e-waste, or consumption of contaminated food and water, can result in various health problems, including respiratory issues, neurological disorders, and cancer.

Resource Depletion: Valuable materials like gold, silver, copper, and rare earth metals are lost when electronic devices are not recycled, contributing to resource scarcity and the need for more energy-intensive extraction processes.

Digital Divide: Improper e-waste management exacerbates the digital divide by hindering access to affordable electronic devices in developing countries, where informal recycling methods often prevail, posing health risks to workers and nearby communities.

Control Measures for E-Waste Management:

Legislation and Regulation: Governments can enact and enforce laws and regulations to promote responsible e-waste management, including bans on hazardous materials, mandatory recycling programs, and extended producer responsibility (EPR) schemes.

Public Awareness and Education: Educating consumers about the environmental and health impacts of e-waste, as well as the importance of proper disposal and recycling, can help change behavior and promote more sustainable consumption patterns.

Collection and Recycling Infrastructure: Establishing convenient collection points and recycling facilities encourages individuals and businesses to dispose of e-waste properly. Improving access to recycling services and incentivizing participation can increase recycling rates.

Technological Innovation: Developing environmentally friendly design principles, such as modular and easily repairable electronics, can extend product lifespans and facilitate recycling.

Additionally, investing in research and development of new recycling technologies can improve efficiency and resource recovery.

International Cooperation: E-waste is a global issue that requires cooperation among countries, manufacturers, and other stakeholders. International agreements and partnerships can facilitate the sharing of best practices, technology transfer, and capacity building to address e-waste management challenges effectively.

OCCUPATIONAL HEALTH AND SAFETY MANAGEMENT SYSTEM (OHSMS)

- An Occupational Health and Safety Management System (OHSMS) is a framework that organizations use to proactively manage and improve workplace safety and health, minimizing risks and preventing accidents and illnesses.
- It's a structured approach encompassing policies, procedures, and processes to ensure a safe and healthy work environment.

Key components:

- **Policy:** Establishes the organization's commitment to health and safety and sets the overall direction for the OHASMS.
- **Planning:** Involves identifying hazards, assessing risks, setting objectives, and developing plans to control hazards and minimize risks.
- **Implementation and operation:** Includes establishing procedures, providing training, ensuring resources are available, and communicating effectively about health and safety matters.
- **Checking and corrective action:** Involves monitoring performance, conducting audits, investigating incidents, and taking corrective actions to address identified issues.

Management review: Ensures the OHASMS is effective and suitable for the organization by reviewing its performance and making necessary changes.

Benefits of an OHASMS:

- Reduced workplace injuries and illnesses
- Improved employee well-being and morale
- Cost savings by lower insurance premiums, reduced absenteeism etc.
- Enhanced reputation among employees, customers, and stakeholders.

Examples of OHASMS elements: Tata Steel in Jamshedpur successfully implemented OHSAS 18001 (now ISO 45001) to improve workplace safety. The company introduced a Safety Management System, conducted Hazard Identification and Risk Assessment (HIRA), and launched a "Zero Harm" safety campaign with regular training for workers.

1. Safety training
2. Emergency preparedness
3. Hazard identification and risk assessment
4. Incident investigation to identify root causes of accidents & prevent recurrence.

Regular safety inspections: As a result, workplace accidents reduced significantly, safety culture improved, and the company received recognition for its strong safety performance.

By implementing an effective OHASMS, organizations can create a safer and healthier workplace, improve their overall performance, and demonstrate their commitment to the well-being of their employees, according to BSI.

Case Studies on OHASMS

Case Study 1: Bhopal Gas Tragedy (India, 1984) – A Case of OHSMS Failure

- In 1984, a massive gas leak occurred at the Union Carbide pesticide plant in Bhopal, India, releasing methyl isocyanate (MIC).
- Over 3,000 people died immediately, and more than 500,000 were exposed to toxic

gas.

Issues in OHSMS:

- Lack of proper safety systems (no functional refrigeration unit to control gas temperature).
- Inadequate emergency preparedness and response plan.
- Workers were not trained in handling hazardous chemicals.
- Negligence in maintenance and monitoring of safety equipment.

Lesson Learned:

- Highlighted the importance of implementing ISO 45001 / OHSAS 18001 type systems for hazard identification, risk assessment, and emergency response.
- Regular audits, safety drills, and employee training are crucial in industries handling toxic materials.

Case Study 2: Tata Steel, Jamshedpur (India) – Successful OHSMS Implementation

- Tata Steel is one of the first companies in India to implement OHSAS 18001 (now ISO 45001) to ensure workplace safety.

Key Initiatives:

- Established a Safety Management System (SMS) across all departments.
- Conducted Hazard Identification and Risk Assessment (HIRA) at each stage of operations.
- Launched a “Zero Harm” campaign and safety training programs for workers. Introduced automation and technology (e.g., drones for plant inspection, wearable devices for workers).

Results:

- Significant reduction in Lost Time Injury Frequency Rate (LTIFR).
- Recognition by the British Safety Council for outstanding safety performance.
- Improved safety culture among employees, contractors, and stakeholders.

Case Study 3: Toyota Motor Corporation (Japan) – OHSMS in Automotive Industry

- Toyota integrated Occupational Health and Safety Management into its Toyota Production System (TPS).

Key Practices:

- Workers encouraged using “Stop-Call-Wait” system: stop the process if a hazard is detected.

- Ergonomic workplace design to reduce musculoskeletal disorders.

- Regular Kaizen (continuous improvement) initiatives focused on workplace safety.
- ISO 45001 certification across global plants.

Results:

- Reduction in workplace accidents and injuries.
- Increased worker morale and productivity.
- Safety improvements also led to higher quality in production output.

Conclusion

- Bhopal Gas Tragedy shows the catastrophic consequences of neglecting OHSMS.
- Tata Steel and Toyota prove that structured occupational health and safety management systems improve safety culture, reduce risks, and enhance productivity.

Environmental protection acts

In India, there are several key environmental protections acts and regulations aimed at safeguarding the country's diverse ecosystems, reducing pollution, and promoting sustainable development. Some of the significant ones include:

The Water (Prevention and Control of Pollution) Act, 1974: This act was enacted to prevent and control water pollution and to maintain or restore the wholesomeness of Water it provides for the establishment of Central and State Pollution Control Boards to enforce its provisions.

Objectives of the Act:

- The Act provides for maintenance and restoration of quality of all types of surface and ground water.
- The Act defines terms like pollution, sewage effluent, trade effluent, stream and boards
- It provides for the establishment of Central and State Boards for pollution control.
- The Act assigns powers and functions to these Boards to control pollution.
- The Central and State Pollution Control Board coordinates and provides technical assistance for prevention and control of water pollution.

- The Act prohibits disposal of any poisonous, lethal or polluting matter to the flow of water in a stream. However, dumping of any material into a stream for the purpose of reclamation of land is not considered an offence.

Draw backs :

- The power & authority has been given only to central government with little of power to state government.
- The penalty is very small when compared to the damage caused by the big industries due to pollution.
- Litigation, related to environment is expensive, since it involves technical knowledge.
- It is very expensive to install Effluent Treatment Plant for small units. Hence violation of law exits.
- The position of chairman of the Board is occupied by political appointee. Hence it is difficult to implement the act without political interference.

The Air (Prevention and Control of Pollution) Act, 1981: Similar to the Water Act, this legislation addresses air pollution concerns. It empowers central and state pollution control boards to regulate and enforce measures to prevent and control air pollution.

Objectives of the Act:

- Provide means for the prevention, control and abatement of air pollution in order to preserve the quality of air.
- The Act defines relevant terms such as air pollution, air pollutants, vehicular exhausts and industrial plants etc.
- The Act also includes automobiles, diesel vehicles, transport, railways and domestic fuels.
- This act can declare certain heavily polluted areas as Air pollution control area and no industrial plant shall be operated in these areas without prior consent of the State Pollution Control Board.
- The State Boards have to lay down and enforce standards for prevention and control of air pollution.
- It ensures emission standards for automobiles.
- The Boards have powers to authorise any person to inspect to collect samples of emissions from chimneys or any other outlets for the analysis of pollutants.
- The Act also includes noise under the category of air pollutants in 1987.

Draw backs:

- Failure to obey the conditions prescribed by the CBCB or SBCB are punishable with fine and imprisonment.
- The State Boards have powers to sue a polluter in a court of law to punish him for polluting the air and the expenses incurred by the Board will be recovered from the polluter.

The Environment (Protection) Act, 1986: This comprehensive legislation empowers the central government to take measures to protect and improve the environment. It authorizes the government to set standards for emissions and discharge of pollutants, regulate environmental impact assessments for development projects, and take other necessary steps for environmental protection.

Salient Features of the Act:

- EPA is a land mark legislation to provide a single focus in the country for the protection of environment and to plug the loop holes in the earlier laws.
- To protect & improvement of the environment.
- To prevent hazards to all living creatures & property.
- To maintain harmonious relationship between human & their environment
- To lay down procedures and safe guards for the prevention of accidents
- To regulate the industries or its operation
- Any person who fails to follow the act shall be punishable with imprisonment for up to five years or with fine up to rupees one lack or both.
- If the violation continues, additional fine of rupees five thousand per day may be imposed
- To inspect the site or plant for preventing pollution and to collect samples of air, water, soil or other material from any factory for testing

The Wildlife Protection Act, 1972: Enacted to protect India's wildlife and biodiversity, this act provides for the protection of endangered species, regulation of hunting and trade in wildlife, and the establishment of protected areas such as national parks and wildlife sanctuaries.

Objectives of the Act

- Restriction and prohibition on hunting and trapping wildlife.
- Rehabilitation of endangered and threatened species.

- Preservation of biological diversity by establishing sanctuaries, national parks and biosphere reserves.
- Grant of a special permit to hunt a wildlife for scientific research, scientific management and collection of specimens for zoological gardens, museums etc.
- Regulation of trade in wildlife and national conservation strategy
- Collaboration with voluntary bodies and NGO's .

Drawbacks:

- It has not included any locally evolved conservation measures
- The personal ownership certificates for animals (tiger, leopard skins etc.) are permissible which is tool for illegal trading.
- The wildlife traders in Jammu and Kashmir get illegal furs and skins from other states. are sold to other countries.
- This is so happening because Jammu and Kashmir has its own Wildlife Act and it does not follow the Central Wildlife Act.
- There is little emphasis on protection of plant genetic resources.
- The offender of the Act is not subject to very harsh penalties. It is just up to 3 years imprisonment or a fine of Rs. 25000 or both.

The Forest (Conservation) Act, 1980: This act aims to conserve forests and regulate forest land use. It requires prior approval from the central government for the diversion of forest land for non-forest purposes such as mining, industry, or infrastructure projects.

Objectives of the Act:

- The Act covers all types of forests including reserve forests, protected forests or any forested land irrespective of its ownership
- The National Forest Policy (1980) prohibits State governments for declaring any portion of forests as non-reserved without approval of Central government
- The amended Act (1988) prohibits lease of forest land to anybody other than the government.

Amended Forest Act, 1992:

- Provision for allowing some non-forest activities in forests without cutting trees with prior approval of Central government.

- Mining is a non-forestry activity and prior approval of Central government is mandatory
- Wildlife sanctuaries, National Parks etc. are totally prohibited for any exploration or survey without prior approval of Central government even if no tree felling is involved.

Drawbacks of the Wildlife

- This Act has just transferred the powers from States to Centres to decide the conversion of reserve forest lands to non-forest areas. Thus powers have been centralized at the top
- The Act has failed to attract public support because it has interfered with the human rights of the poor native people.
- Poor participation of the community
- Tribal communities have a rich knowledge about the forest resources, their values and conservation. But their role and contribution is neither acknowledged nor honoured.
- Efforts are now being made to make up for gaps in laws by introducing the principles of Public trust or Human rights protection

The National Green Tribunal Act, 2010: The National Green Tribunal (NGT) is a specialized judicial body established under this act to handle cases related to environmental protection and conservation. It has the authority to hear and dispose of cases involving environmental disputes and violations of environmental laws.

The Coastal Regulation Zone (CRZ) Notification, 2011: This regulation aims to protect and manage India's coastal areas, including beaches, mangroves, and wetlands. It sets restrictions on activities such as construction, industrial development, and mining in coastal areas to prevent ecological damage.

These are some of the key environmental protection acts and regulations in India, although there are many more at both the central and state levels aimed at addressing various environmental issues facing the country.

Part B (Frequently/Previously asked semester questions)

1. Define and write the causes, effects and preventive measures of
 - a) Air pollution b) water pollution c) soil pollution d) noise pollution
2. Explain in detail: Solid waste management.
3. Explain in detail: Hazardous waste and E-waste management.
4. Write notes on various environmental protection acts.
5. Explain the role of an individual for the pollution prevention programme.

Part – A (Frequently/Previously asked semester questions)

1. Give any four major secondary air pollutants that are present in the environment.

PAN (peroxy acetyl nitrate), ozone, formaldehyde, acid mist, etc

2. Mention the A) Point and B) Non – point sources of water pollution?

A) Industrial effluent and sewage B) Excess fertilizers and pesticides from agricultural lands

3. Distinguish between sewage and effluent.

- ✓ Sewage: The waste water released from the domestic, commercial, municipal, hotel, hospital, institutions, etc are called sewage.
- ✓ Industrial effluent: The waste water discharged from various industries like textile, dye, cement, paper is called effluent.

4. What are hazardous wastes?

Give any two major sources of hazardous wastes. The waste which causes potential or substantial threats to public health or the environment is called hazardous waste. Mining waste & nuclear waste are two major sources of hazardous wastes.

5.What is Composting? How does it occur?

The combined process of decomposition and stabilization of solid waste by suitable biochemical bacteriological method are called composting. It occurs through aerobic and anaerobic decomposition.

6.State the role and responsibility of an individual in the prevention of pollution.

- ✓ The population stabilization to be achieved to avoid further environmental pollution.
- ✓ The concept of Reduce, Re-use and Recycle (3 R – Mantra) must be followed.
- ✓ The usage of chemical fertilizers must be banned and natural manure should be allowed.
- ✓ Every individual should educate others about environmental degradation and control, impact through pamphlets, printed folio, etc

7. What are the various sources of radioactive pollution?

- ✓ The leakage of radiations from nuclear reactor, nuclear research laboratories, etc
- ✓ The transportation, disposal and storage of radioactive waste, etc

- ✓ The process of mining and refining of radioactive metals, U, Th, Pt, etc
- ✓ Nuclear test, the radioactive isotopes mixed with atmospheric air and bring pollution.

8. When a sound causes noise pollution?

- 1) Blaring of loudspeaker in public places.
- 2) Bursting of crackers, Road traffic, air craft taking off, massive industries, etc.

9. Define soil pollution.

The degradation of soil and land due to industrial, agricultural and other man-made activities is called soil pollution.

10. What is meant by anthropogenic sources of pollution?

The pollutants discharged into atmosphere by man-made activities are called anthropogenic pollutants. E.g. Oxides of S, C and N, CFC, Ammonia, sewage, etc.

11. What is meant by eutrophication?

The algae bloom in the surface water can produce bad taste, odour and green colour to the fresh water, so that the water is unfit for drinking due to the lack of dissolved oxygen. This process is called eutrophication.

12. What is meant by E – waste?

E – Waste means discarded electrical or electronic devices. It includes discarded computers, office electronic equipment, mobile phones, television and refrigerators, etc.

13. List any two air pollutants and their effects on human beings from automobile emission.

- ✓ Carbon monoxide causes headache, dizziness, vomiting and may be unconscious
- ✓ Hydrocarbon – Giddiness, narcotic effect and cardiac arrest.

14. State the objectives of treatment of waste water.

- ✓ To remove the heavy metal (Fe, Mn, Ni & Co)
- ✓ contamination from industrial effluent
- ✓ To reduce the toxicity present in the effluent and sewage

15. What are the types of solid wastes?

- ✓ Bio-degradable solid waste: The pollutants which are rapidly decomposed then they are called bio- degradable pollutants. Example. Sewage and effluents.
- ✓ Non-bio-degradable solid waste: The pollutants which are not rapidly decomposed then they are called non bio-degradable pollutants. Example. Mercury, plastics, glass, etc.

16. Write any four major water pollutants.

- ✓ Sewage
- ✓ Effluent from textile, dye and paper industries
- ✓ Acid rain
- ✓ Algae and fungi in water bodies

17. What are the causes of air pollution?

- ✓ The toxic substances released atomic and thermal power plant, decay of organic matter, burning of fossil fuels, etc.
- ✓ The CO gas emitted from motor vehicles.
- ✓ Smoke from fire crackers and oil refineries
- ✓ Domestic fire, incineration, etc
- ✓ Natural disaster like forest fire, volcanoes, dust storm, etc.

18. Define Noise pollution.

- ✓ Give any two controlling measures. The excess sound created by human activities is called noise pollution.
- ✓ Source reduction – It is process of stopping the noise before it gets started.
- ✓ Planting trees and shrubs along roads, the hospital, college, industries, etc help in reducing to considerable extent.

19. Give any four major soil pollutants.

- ✓ Bio-medical waste from hospitals
- ✓ Industrial effluents and sewage
- ✓ Insecticides, pesticides and fertilizers, etc
- ✓ Non-biodegradable materials such as plastics, rubber and glass, etc

20. What are the risks associated nuclear power plants?

- ✓ The radiation from nuclear power plants causes health problems
- ✓ The nuclear accidents causes severe damages to biodiversity
- ✓ The nuclear waste disposal to be isolated.
- ✓ The huge investment in installation.

21. What is Photochemical smog?

- ✓ The brownish smoke like appearance that frequently forms on clear, sunny days over large cities with significant amount of automobile traffic. It is mainly due to chemical reactions amount nitrogen oxide and hydrocarbon by sunlight.